

TAMWORTH, Australia

WMO: 957620

Lat: 31.0742S

Lon: 150.8361E

Elev: 396

StdP: 96.66

Time Zone: 10.00 (AUS)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.1	0.3	-5.3	2.5	13.3	-3.8	2.9	10.9	9.4	13.9	8.5	14.1	1.6	140	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	36.1	19.2	34.2	18.9	32.6	18.8	22.2	29.0	21.5	28.5	20.8	27.7	5.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.7	16.2	23.6	19.7	15.1	23.3	18.8	14.2	23.2	67.4	28.7	64.4	28.5	61.9	27.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.5	7.5	DB	-4.1	38.6	1.1	1.7	-4.9	39.8	-5.5	40.8	-6.1	41.8	-6.9	43.0	
			WB	-4.6	23.7	1.2	1.5	-5.4	24.8	-6.2	25.7	-6.8	26.5	-7.7	27.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	24.6	24.0	21.5	17.5	13.2	10.2	9.1	10.6	14.0	17.3	20.4	22.6
	DBStd	6.07	2.84	2.85	2.51	2.71	2.83	2.74	2.28	2.89	3.00	3.08	3.51	3.04
	HDD10.0	112	0	0	0	1	6	31	46	26	3	1	0	0
	HDD18.3	1193	0	1	4	45	161	244	288	239	134	56	17	3
	CDD10.0	2680	452	392	358	225	105	37	16	46	121	226	311	392
	CDD18.3	720	194	159	104	20	1	0	0	1	3	23	78	137
	CDH23.3	8093	2175	1627	984	257	17	0	0	9	109	410	940	1564
	CDH26.7	3395	1069	739	332	34	1	0	0	1	21	114	391	693
(14) Wind	WSAvg	3.5	4.1	3.9	3.7	3.3	3.0	2.9	2.9	3.2	3.4	3.6	3.9	3.9
(15) Precipitation	PrecAvg	676	113	58	44	40	51	35	46	44	45	61	67	69
	PrecMax	946	322	202	145	124	150	98	191	128	122	146	159	152
	PrecMin	359	6	4	0	0	1	2	4	0	2	12	5	3
	PrecStd	147	79	50	38	39	41	26	41	32	27	32	37	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.2	38.0	34.2	29.4	25.2	21.5	19.6	24.4	29.1	32.0	36.8	37.3
		MCWB	20.0	19.9	18.3	16.9	15.4	14.0	12.2	14.4	15.0	16.0	17.9	18.6
	2%	DB	36.1	35.1	31.8	27.6	23.2	19.5	17.9	21.8	26.3	29.6	33.5	34.4
		MCWB	19.1	20.2	18.2	16.6	14.3	12.9	10.9	12.7	14.6	15.5	17.5	18.4
	5%	DB	34.3	33.0	30.1	26.2	21.7	18.0	16.8	19.9	24.1	27.6	31.0	32.6
		MCWB	19.1	19.8	17.9	16.0	13.5	12.0	10.7	11.9	13.9	15.4	17.6	18.4
	10%	DB	32.4	31.1	28.4	24.7	20.2	16.6	15.3	18.0	22.0	25.5	28.6	30.7
		MCWB	19.2	19.3	17.7	15.5	12.9	11.2	10.2	10.9	13.5	15.0	16.8	17.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	23.5	23.3	23.8	17.3	16.0	14.7	15.6	17.6	19.7	21.1	22.2
		MCDB	31.2	30.6	24.3	23.8	21.6	19.5	16.3	22.1	23.5	26.4	29.2	29.7
	2%	WB	21.7	22.4	21.3	19.0	16.1	14.3	13.1	14.2	16.4	18.2	19.9	21.0
		MCDB	29.6	29.3	26.4	23.3	20.5	17.2	15.5	18.9	22.7	24.3	27.6	28.3
	5%	WB	21.1	21.6	20.2	17.5	15.0	13.1	11.8	13.0	15.4	17.1	19.0	20.3
		MCDB	28.8	28.5	25.7	22.8	19.0	16.1	15.2	18.1	21.2	23.4	26.1	27.5
	10%	WB	20.5	20.8	19.2	16.4	13.9	12.1	10.7	11.9	14.2	16.1	18.2	19.6
		MCDB	28.0	27.5	25.3	22.5	18.3	15.7	14.4	17.0	20.1	22.5	25.1	26.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	13.1	13.4	14.1	13.7	12.5	12.7	14.4	15.0	14.7	13.8	13.9
		MCDBR	16.9	15.9	15.6	15.8	15.5	14.2	14.9	17.0	18.3	18.3	17.8	16.8
	5% WB	MCWBR	4.8	4.6	5.1	6.6	7.4	8.1	8.8	9.0	8.3	7.6	5.7	5.3
		MCDWR	12.8	12.5	12.0	12.8	11.2	10.9	11.4	13.5	14.5	13.9	13.0	12.9
(40) Clear-Sky Solar Irradiance	taub		0.356	0.352	0.326	0.308	0.288	0.283	0.278	0.290	0.323	0.337	0.358	0.354
		taud	2.516	2.546	2.609	2.620	2.623	2.645	2.639	2.584	2.478	2.474	2.467	2.505
	Ebn at Noon		984	969	962	931	904	886	907	937	949	972	975	989
		Edh at Noon	113	106	94	83	74	68	71	84	105	113	118	115
(44) All-Sky Solar Radiation	RadAvg		7.14	6.42	5.42	4.45	3.48	2.77	3.16	4.11	5.37	6.24	6.69	7.15
	RadStd		0.47	0.52	0.42	0.29	0.23	0.23	0.22	0.28	0.39	0.54	0.68	0.55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	-24	-130	+190

Nomenclature: See separate page